

Video Game Programming: Serious Games

L23

Prof. Fabio Buttussi, PhD
Human-Computer Interaction Laboratory
Department of Mathematics, Computer Science, and Physics
University of Udine - Italy

Serious game research

- The idea of using games for serious purposes is increasingly attracting academic interest
- Involved researchers include computer scientist, psychologists, and educators
- The number of yearly research publications grew from 42 in 2007 to 237 in 2017
- Besides specific conferences and journals, major higher-ranked journals (e.g., C&E, IJHCS, TVCG, CHB) regularly publish articles about serious games

Serious game market

- On one hand, the wide spread of commercial, entertainment games supported efforts on serious games
- On the other hand, serious games helped improving the perception of games by showing how they can be more than FPS for teenagers
- The barriers for larger use in schools, corporations, government are slowly disappearing
- Market research companies like Gartner, IDATE, and Allied Market Research predict a very larger serious game market
- E.g., Allied Market Research predicts growth from \$2,731 million in 2016 to \$9,167 million in 2023

Motivation and game mechanics

- **Motivation** can be extrinsic or intrinsic:
 - **Extrinsic** motivation draws on game elements that are external and ad hoc to the gameplay (e.g., points, leader board, or level completion), which can be used across different gameplays
 - **Intrinsic** motivation relies on the gameplay by itself to motivate and engage, and do not necessarily need external motivation elements
- **Game mechanics** are exogenous or endogenous:
 - With **endogenous** (internal) game mechanics, learning elements are directly embedded in the gameplay as part of in players' actions
 - With **exogenous** (external) game mechanics, there is not such a direct relationship

Learning theories

- Serious games should be based on solid learning theories
- Behaviorism:
 - It set educational game challenge in transmitting information from the game to the player, considering information context less relevant
 - The transmission of content relies on conditioning and reinforcement
 - Behaviorism-based games usually rely heavily on simple game mechanics
- Constructivism:
 - It promotes players' freedom to explore the VE and the process of constructing knowledge in a meaningful and personal way
 - Games should make it possible for learners to approach a subject in an active way and construct their own representations
 - Constructivism-based game should draw on different perspectives, give rise to a variety of actions, and offer a full understanding of the given topic

Serious games before video games

- Before the adoption of the term, the idea of using games for serious purposes was developed using different names
- First educational games in the late '50s were non-electronic, pen-and-paper and board games
- By the '70s, educational games exploded in popularity, and became important pedagogical tools
- Most of them were simulations, and they were especially used for teaching in American businesses and the military
- E.g., in Inter-Nation Simulation (1958), high-school players controlled hypothetical nations that should negotiate

Early serious game initiatives

- With the spread of video games, they started to be used especially in school, military, health, or corporate context
- The Oregon Trail (1971, in fig.) was one the first learning games and is considered one of the greatest games ever
- Born as a text-based strategy game, it later became a graphical adventure
- The goal is to take care of the needs of 5 emigrants on the trail



Games for change

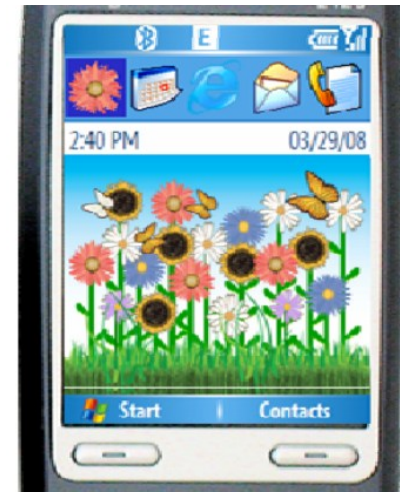
- Games for change focus on making the world a better place by teaching positive behaviors and supporting behavior change
- McGonigal (in Reality Is Broken) suggested that games can serve as backdrop for serious collaboration, thinking, and reflection
- She introduced the term alternate reality game (ARG), weaving together fiction and reality to engage people in solving real-life problems in an epic setting
- E.g., in World Without Oil, players should try to engineer, develop, and build solutions in a world with an oil shortage

Games for health

- With Games for health conference and journal, these kind of games was one of the first major contributors to serious games
- In a review study from 2018, 1,738 games for health were identified (Lu and Kharrazi, A State-of-the-Art Systematic Content Analysis of Games for Health, Games for Health Journal, 7:1, 1-15, 2018)
- Games for health can involve staff, patients, and health people
- Games can be used for treatment, rehabilitation, compliance, and training purposes
- They can support knowledge and skill gain as well as behavior change

Exergames

- Both academics and industry devoted considerable attention to exergames, which require players to perform physical exercise
- Among exergames, fitness games support players performing fitness exercises and were largely used to combat obesity
- They rely on mobile device sensors (e.g., GPS, accelerometer) or new peripherals for PCs and consoles (e.g., Wiimote, balance board, Eye Toy, Kinect)



Consolvo et al., Flowers or a robot army?
Encouraging awareness & activity with personal,
mobile displays, Proc 10th Intl Conf Ubiquitous
computing, 54-63, 2008



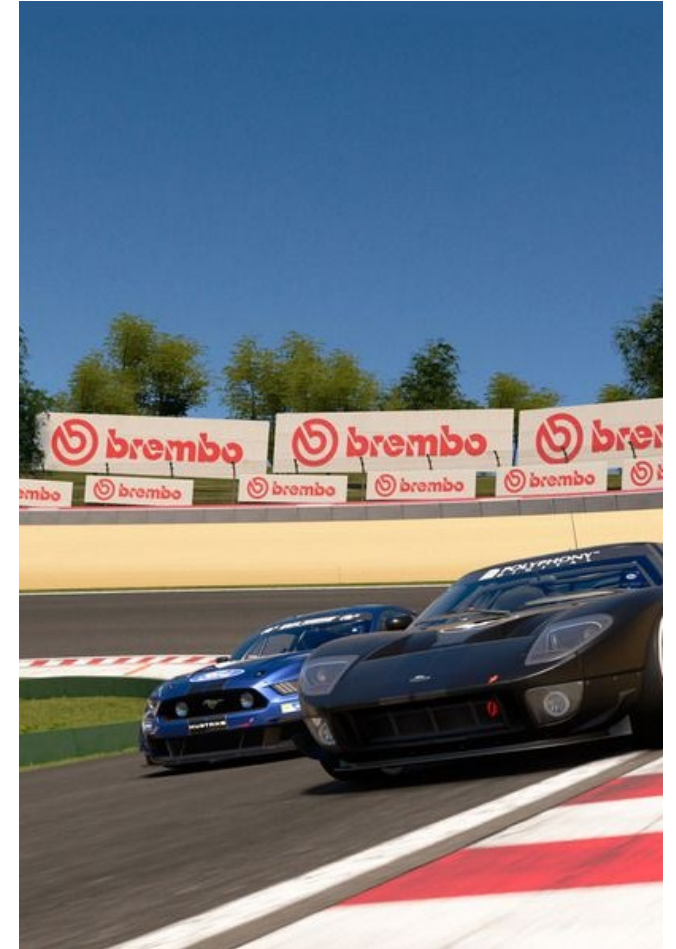
EA Sports Active 2

News & political games

- In news games, players are involved in news stories
- The genre experienced some popularity in 2010-2011 with games like Playing the News and Cutthroat Capitalism
- Political games aim at pushing a political message to the player by putting them in a specific position
- Early examples from the '80s are Nuclear War, Balance of Power, Hidden Agenda
- The most famous is America's Army, which is explicitly offered as a promotional tool to inspire youth to join the US Army

Advergames

- Advertainment: fusion of advertising and entertainment, refers to games for marketing
- Grew considerably from '90s up to late 2000s
- E.g., Three King Games, made by a fast-food franchise in 2006, was sold on Xbox store
- LEGO uses online and mobile games to build brand
- Products are incorporated into the gameplay in creative(?) ways (integral games vs. giveaways)
- Original motivation (“build it and they will come”) is no longer valid because free games are available
- A new way to reach the same audience are e-Sports



Ads in GT7

Edutainment

- Edutainment focuses on teaching skills, e.g., algebra, spelling
- It started in the '70s, typically targets children, and basically relies on behaviorism learning principles
- Examples include Seven Cities of Gold, about Spanish colonization of Latin America, and Math Blaster, where players shoot down answers to arithmetical problems



Seven Cities of Gold



Math Blaster

Edutainment limitations

- Gameplay and graphics are usually considered dated and simplistic with respect to commercial games
- Edutainment has a strong educational component, but often does not support engagement, curiosity and exploration
- Drill-and-practice learning rather than understanding, e.g., games require to memorize answers, but do not teach underlying rules
- Use of extrinsic motivation based on arbitrary rewards rather than intrinsic motivation arising from the game activity per se
- Use of exogenous game mechanics with limited integration with learning activities, so players may concentrate on playing rather than learning
- Limited player agency, with players often not able to do anything actively with the game content they are learning

Commercial off-the-shelf games

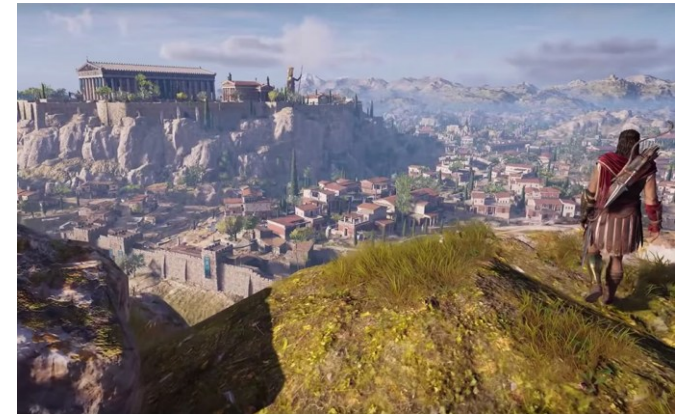
- Because of limited edutainment success, researchers and educators tried using commercial off-the-shelf (COTS) games
- COTS games rarely focus on teaching topics or skills, but some can be remapped to a curriculum
- Games as SimCity, Civilization, The Sims, and Minecraft have been used by numerous schools
- E.g., in SimCity, players must plan and run a small city by understanding basic principles of urban planning
- Motivational effect is documented by their commercial success

COTS game pros and cons

- COTS games support explorative learning, but principles are often indirect or tangential
- COTS games support intrinsic motivation thanks to their core gameplay
- Player agency is strong because players are involved in the game experience, but focus may be more on game than learning
- COTS games heavily rely on teachers' support for learning, e.g., a debriefing phase may correct mistakes, clarify misconceptions, and expand on the game experience



Hearts of Iron IV



Assassin's Creed Odyssey

Serious Games

- Serious games (SGs) try to combine the pros of COTS games with the strong educational component of edutainment
- Serious games (SGs) share the desire for intrinsic motivation and endogenous design coupled with a strong player agency, where players actively use the learning as part of the game experience
- SGs should thus integrate gameplay and educational content
- Edutainment was mostly constrained by limited resources
- Chances of developing better SGs have increased thanks to the availability of game engines and assets

SG potential

- Fantasy: the SG activity should increase intrinsic motivation by using internal or external fantasies as a part of the game world
- Control: the player should gain the feeling of being in control
- Challenge: the activity should be at the appropriate level of difficulty for players to be pushed to the limit of their capacity
- Curiosity: information in the SGs should be complex and unknown to encourage exploration and reorganization of the information
- Situated learning: SGs are well suited for new forms of learning in which players are situated in the domain and understands it from the inside

SG research goals

- SG research is often aimed at documenting positive outcomes
- It addresses different target groups, e.g., children, youth, and adults
- It addressed both theoretical knowledge and practical skills
- Key research questions concern effectiveness, motivation, debriefing, and influence of teachers and setting
- In general, SGs proved to be a viable alternative to traditional teaching, and provide approximately the same learning outcome
- SGs cannot necessarily be said to be more effective than other teaching methods, but most studies have offered evidence of higher motivation, better retention over time and potentially better transfer

SG surveys

- Overall, surveys on SGs suggest that:
 - SG type influence learning outcomes
 - Theoretical foundations hold up in studies
 - SGs informed by learning theories perform better, but this does not imply complicated game mechanics

Reference	Genre	Studies Participants		Results (+ indicates positive impact)
Vogel et al. (2006)	Computer games and interactive simulations	32	All age groups	+ learning outcome in cognitive + students' motivation + player control game
Sitzmann and Ely (2010)	Simulation games	65	+18 years	+ declarative and procedural learning + higher self-efficacy + player active + higher retention and better transfer
Connolly, Boyle, MacArthur, Hainey, and Boyle (2012)	Computer games and serious game	120	+14 years	+ Learning outcomes in all area except soft skills
Wouters and Oostendorp (2013)	Serious games	39	All age groups	+ Improvement in knowledge + Improvements in skills + Instructional support + Collaborative games
Clark et al. (2013)	Digital games	69	K-16 students	+ Learning outcomes + Noncompetitive games + Learning theory informed games
Lamb, Annetta, Firestone, and Etopio (2018)	Serious educational games, educational simulations and serious games	46	Secondary to university	+ Learning outcome in cognitive, affective and skills

Research challenges in SGs

- Learning vs. Playing
- Drill-and-Practice vs. Microworlds
- Immersion vs. Transfer
- Teacher Role and Intervention
- Assessment

Gamification

- Gamification is the use of game design elements in non-game contexts (Deterding et al., From Game Design Elements to Gamefulness: Defining “Gamification”, Procs 15th Intl Ac MindTrek Conf Envisioning Future Media Environments, 9-15, 2011)
- Gamification brings game into the activity, rather than the opposite
- The instrumentalist approach delivers the “how” of behavioral change through gaming, without worrying too much about the simple “why” :
 - Nike+, where players receive points by doing physical exercise
 - Foursquare, where players receive points by checking-in places
- The idealist approach stresses that the gamified activity should not only get people to do things but also provide a deeper, meaningful activity:
 - Chore Wars, where players can set up their own household activities, points, and goals, maintaining freedom and autonomy
 - Toyota Prius, where a green plant in the dashboard shows how green you are driving, allowing players to create their own challenges and goals

Key elements

- Key gamification elements were introduced in Deterding et al., From Game Design Elements to Gamefulness: Defining “Gamification”, Procs 15th Intl Ac MindTrek Conf Envisioning Future Media Environments, 9-15, 2011
- Several researchers argue that more game elements should be added

<i>Level</i>	<i>Description</i>	<i>Example</i>
Game interface design patterns	Common, successful interaction design components and design solutions for a known problem in a context, including prototypical implementations	Badge, leaderboard, level
Game design patterns and mechanics	Commonly reoccurring parts of the design of a game that concern gameplay	Time constraint, limited resources, turns
Game design principles and heuristics	Evaluative guidelines to approach a design problem or analyze a given design solution	Enduring play, clear goals, variety of game styles
Game models	Conceptual models of the components of games or game experience	MDA; challenge, fantasy, curiosity; game design atoms; CEGE
Game design methods	Game design-specific practices and processes	Playtesting, playcentric design, value conscious game design

SGs vs. gamification definition

- Serious games (product): The concept that video games can be developed for and used in a variety of non-entertainment settings
- Gamification (process): The concept that game mechanics can be used in a variety of non-entertainment settings
- Serious games typically simulate or represent something to support players' learning in a virtual environment
- Gamification is a layer on top of real life to change the way players learn, behave, or think

SGs vs. gamification examples

- In Dragon Box (SG, in fig.), players learn about algebra by playing a game, where they must solve several puzzles; the game introduces key concepts related to algebra that players need to apply successfully to progress in the game
- In Sales Force Motivation (gamification), the sales staff is challenged to get points on the score board by selling; through the leaderboard salespeople are constantly engaged and motivated to sell the most to beat their colleagues



SGs vs. gamification comparison

	<i>Serious Games</i>	<i>Gamification</i>
Alternative names	Edutainment, learning games, game-based learning, educational games, simulation games	Gameful design, playful design, motivational design, ludification
Scientific and theoretical starting point	Educational theories, instructional design, education psychology, simulation theory	Behavioral design, behavioral economics, game theory, social psychology, nudging
Focus	Product	Process
Player actions	Virtual	Real
Center	Individual	Group
Game role	Simulation/representation	Layer/tracker/tool
Purpose	Create learning	Change behavior

Edutainment vs. COTS games vs SGs

	<i>Edutainment</i>	<i>Commercial off-the-shelf games (COTS)</i>	<i>Educational serious games</i>
Game learning focus	Skills and knowledge	None or Knowledge	Skills, knowledge and affective
Instructional theory	Behavioristic	Common sense	Behavioristic, cognitive, constructivism
Motivation	Extrinsic	Intrinsic	Intrinsic
Game mechanics	Exogenous	Mixed	Endogenous
Player agency	Passive	Active	Active
Teacher role	None	Ad hoc	Active
Debriefing	None	Ad hoc	Integrated
Game examples	<i>Math Blaster, Monkey Math School</i>	<i>Minecraft, Civilization, SimCity</i>	<i>Oregon Trail, Dragon Box</i>

Bibliography

- Egenfeldt-Nielsen, Smith, Tosca, Understanding Video Games: The Essential Introduction, 5th ed., Routledge, 2024

